

Real-Time 6D pose estimation using FoundationPose

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Introduction

Robotic manipulation plays a crucial role in industrial automation, logistics, and service robotics. To enable efficient and autonomous manipulation, robots must be capable of accurately detecting, localizing, and grasping objects in real-time. A critical component of this process is 6D pose estimation, which refers to the task of determining the 3D position (X, Y, Z) and 3D orientation (roll, pitch, yaw) of an object relative to a camera. This process is important for tasks including object manipulation, navigation, and interaction with the environment. In robotics, it enables robots to manipulate objects with high precision, making them essential for automation tasks. In the context of pick-and-place tasks, precise estimation of 6D poses ensures that the robot can align its gripper accurately with the object. 6D pose estimation is also widely used in autonomous systems, such as self-driving vehicles and drones, which allows the system to understand the position and movement of objects in their surroundings, ensuring safe navigation. In augmented reality, 6D pose estimation improves user experiences by placing virtual objects correctly in the real world, creating seamless interactions between digital and physical elements.

While many existing approaches focused on synthetic dataset, they often struggle with unseen objects and noisy rgb-d inputs. Models like PoseCNN and DendeFusion are trained such that each object has its own weights and model associated with it. As a result, these models do not generalize to new objects without retraining. PVNet on the other hand by predicting the 2d key points instead of the 6d pose directly. Rather than directly predict the key points, which might be occluded, it lets other pixels generate a vector that points to the potential keypoints, and then use RANSAC to estimate the keypoints. This makes PVNet largely more robust in occasions with occlusion. Foundation Pose brings another level of generalization using semantic features learned from CLIP, which improves the ability to handle unseen objects and real time inferences.

Our project focuses on implementing a real-time 6d pose estimation pipeline using FoundationPose. We used ArUco gridboard for RGB-D camera calibration and built the model with a 3d printed LINEMOD gorilla. We then test the pose estimation across both static and dynamic environments and evaluate the robustness.

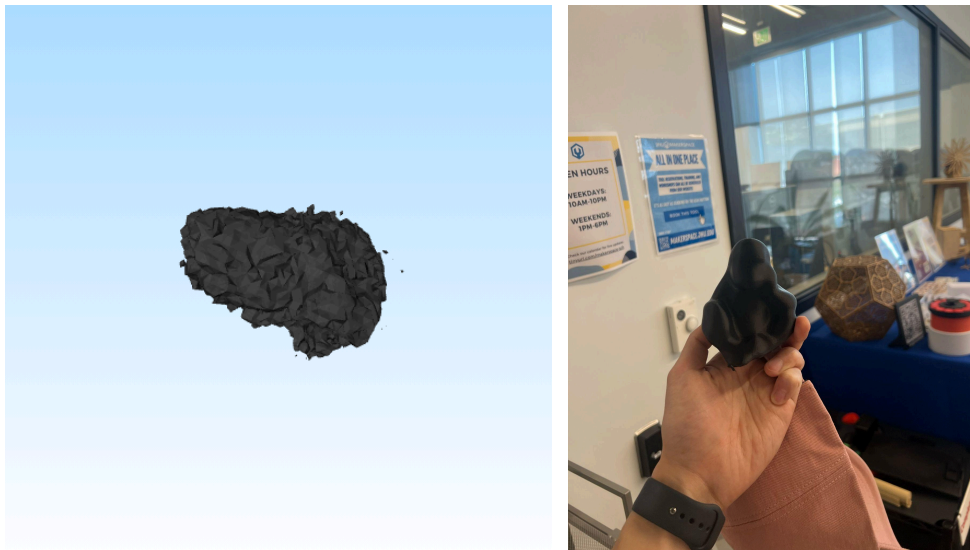
Method

We use NVIDIA's FoundationPose model which outputs six-degree-of-freedom object pose estimation and tracking without any task-specific fine-tuning. At its core, FoundationPose combines a transformer-based pose refinement network with a pose selection module and, optionally, a neural object field for implicit surface representation. To locate objects accurately,

we used SAM to generate the object mask from the RGB input. These masks are used along with the RGB data, aligned depth data and 3D object mesh to provide complete contextual information for accurate pose estimation. Together, these components enable reliable orientation and translation estimates directly from RGB-D inputs.

Accurate camera calibration is critical to our pipeline. We printed a custom ArUco gridboard on standard letter-size paper, arranging 20 square markers (each 28 mm per side, with 4 mm spacing) in a 5 by 7 grid around the sheet's perimeter using OpenCV's DICT_6X6_250 dictionary. By capturing multiple viewpoints of this board and detecting its markers in the RGB channel, we recover both intrinsics and extrinsics parameters of the camera, along with the RealSense SDK's RGB-to-depth offsets to get a full world-to-depth transformation. We retrieve intrinsic parameters for color and depth using the software and apply its on-chip self-calibration to refine any residual distortion or alignment errors.

To validate our setup, we 3D-printed the gorilla mesh from the LINEMOD dataset and used its CAD geometry to initialize FoundationPose's model-based tracker. The live video feed then confirms real-time tracking performance on this physical prop. We also explored a NeRF-style reconstruction from our multi-view RGB captures, but found that limited depth resolution and sparse viewpoints prevented the implicit model from constructing a high fidelity object model.



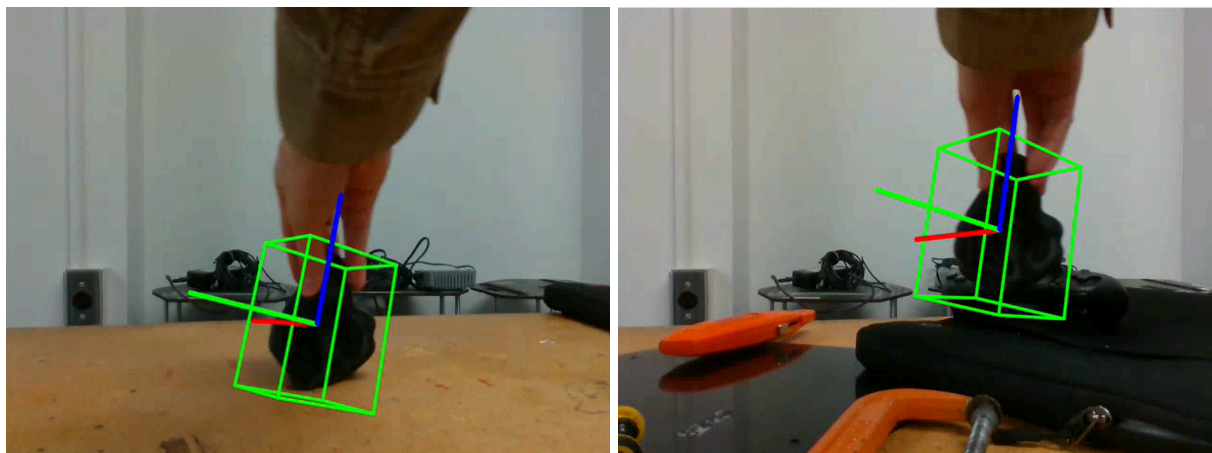
NeRF generated 3D model of the Gorilla

Results

We began by using the LINEMOD dataset to run the FoundationPose model. NeRF reconstruction was applied to the dataset to generate CAD geometries for 15 different objects. We then ran the FoundationPose model on each object, and the results were promising, which encouraged us to move on to real-time testing with a custom dataset.

Focusing on the "Gorilla" model from the LINEMOD dataset, we converted its 3D geometry into a format suitable for 3D printing and produced a physical replica. By parsing video footage into image frames and extracting the mask from the first frame, we were able to run the FoundationPose model in real time. The results were surprisingly accurate, particularly when the camera was fixed. We successfully visualized and obtained both the orientation and translation of the target object in real time.

The performance of the pose estimation varied depending on the environment. It remained robust in both clear and cluttered scenes, accurately detecting fast-moving or rotating objects and those with backgrounds of similar color. However, performance degraded when the camera moved along with the object, leading to drift in the bounding box and incorrect orientation estimates.



Pose Estimation of the Gorilla in clear and messy background

Discussion

We tested real-time 6D pose estimation using various datasets, environments, and 3D objects. Our initial goal was to implement 6D pose estimation and then integrate grasp planning and KUKA arm control methods from a parallel project led by colleagues. However, due to unforeseen challenges in that project, we were unable to complete the integration with the physical robot in time. Despite this, the FoundationPose model performed reliably as intended. It successfully detected the target object and provided accurate orientation and translation data—even when the object was rotating or translating at relatively high speeds.

One key limitation we faced was the difficulty in quantitatively evaluating our results. Since we lacked ground truth pose data for comparison, it was not possible to benchmark our model's accuracy against other approaches. Furthermore, given that FoundationPose is a recently published method (2024), comparative baselines or established benchmarks were not readily available.

Looking ahead, several directions can enhance this work and contribute meaningfully to the field:

1. **Multi-object Pose Estimation:** Extending the system to handle multiple objects simultaneously would enable selective object manipulation, which has direct applications in warehouse and logistics automation.
2. **Custom Dataset Training:** By collecting our own data and using NeRF to generate 3D reconstructions, we could train the FoundationPose model on objects of interest, leading to more robust and task-specific performance.
3. **Robot Arm Integration:** As originally intended, integrating motion planning with a robot arm would enable autonomous grasping of objects with arbitrary orientations and positions, paving the way for real-world robotic manipulation.

Conclusion

In this project, we successfully implemented a real-time 6D pose estimation pipeline using the FoundationPose model, demonstrating its robustness in both static and dynamic environments. By combining accurate camera calibration, custom 3D-printed objects, and RGB-D data processing, we were able to localize and track objects with impressive accuracy and stability. Although integration with the robotic arm was not completed due to external project dependencies, the system showed strong potential for future deployment in autonomous manipulation tasks. Moving forward, extending the approach to multi-object scenarios, incorporating custom training data, and achieving full robotic integration will further advance the applicability of this work in real-world robotics and automation systems.

Contributions

- **Image Segmentation (Mask Generation using SAM):** Rayna, Ryan
- **RGB-D RealSense Camera (Calibration and Image Alignment):** Sungwoo
- **Custom Dataset Construction:** Rayna, Ryan, Sungwoo
- **3D Modeling and Printing:** Sungwoo
- **Report Writing:** Sungwoo, Rayna, Ryan

Citation

Xiang, Yu, et al. "PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes." arXiv, 1 Nov. 2017, revised 26 May 2018, <https://doi.org/10.48550/arXiv.1711.00199>.

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Wen, Bowen, et al. "FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects." 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), vol. 35, June 2024, pp. 17868–79. <https://doi.org/10.1109/cvpr52733.2024.01692>.

NVlabs. "GitHub - NVlabs/FoundationPose: [CVPR 2024 Highlight] FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects." GitHub, github.com/NVlabs/FoundationPose.